

S.Y.B.Sc.(Data Science)

Sem- IV

Subject: Data Structure-II

Unit1 : Tree

By,

Mrs. Reshma Masurekar

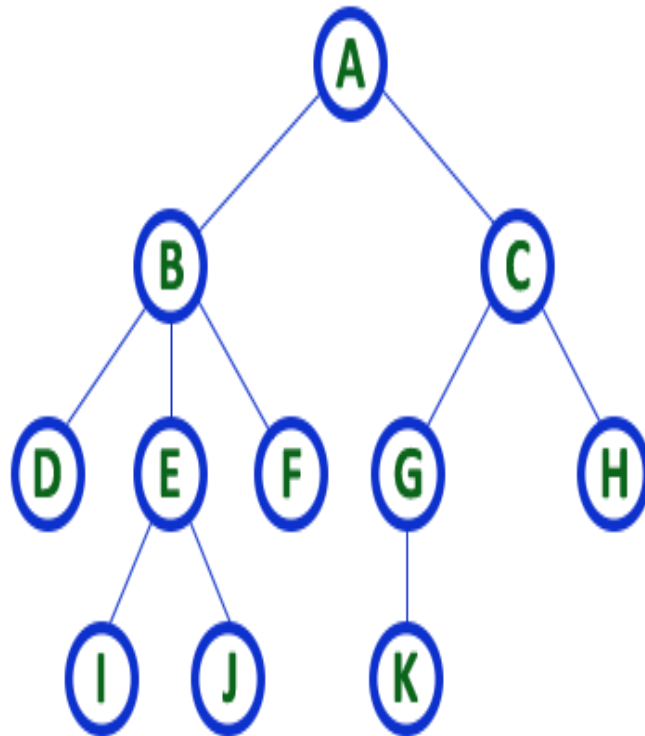
***Dr. D. Y. Patil Arts, Commerce and Science College,
Pimpri, Pune 18***

Chapter 1:Tree

- **Concept and Terminologies**
 - Types of Binary Tree-Binary Tree, Skewed tree, Strict Binary Tree, Full Binary Tree, Complete Binary Tree, Expression tree, Binary Search Tree, Heap
- **Representation**
 - Static and Dynamic Representation of tree
- **Implementation and Operations on Binary Search Tree**
 - Create, Insert, Delete, Search
 - Tree Traversal-preorder, inorder, postorder ,Level order traversal using queue
 - Counting Leaf node, Non leaf nodes, Copy, Mirror
- **Application of Trees**
 - Heap Sort
 - Introduction to Greedy Strategy, Huffman encoding (Implementation Using Priority Queue)
 - **Applications of tree in Data Science**
 - Decision tree in machine learning
 - Hierarchical clustering in data analysis

Definition

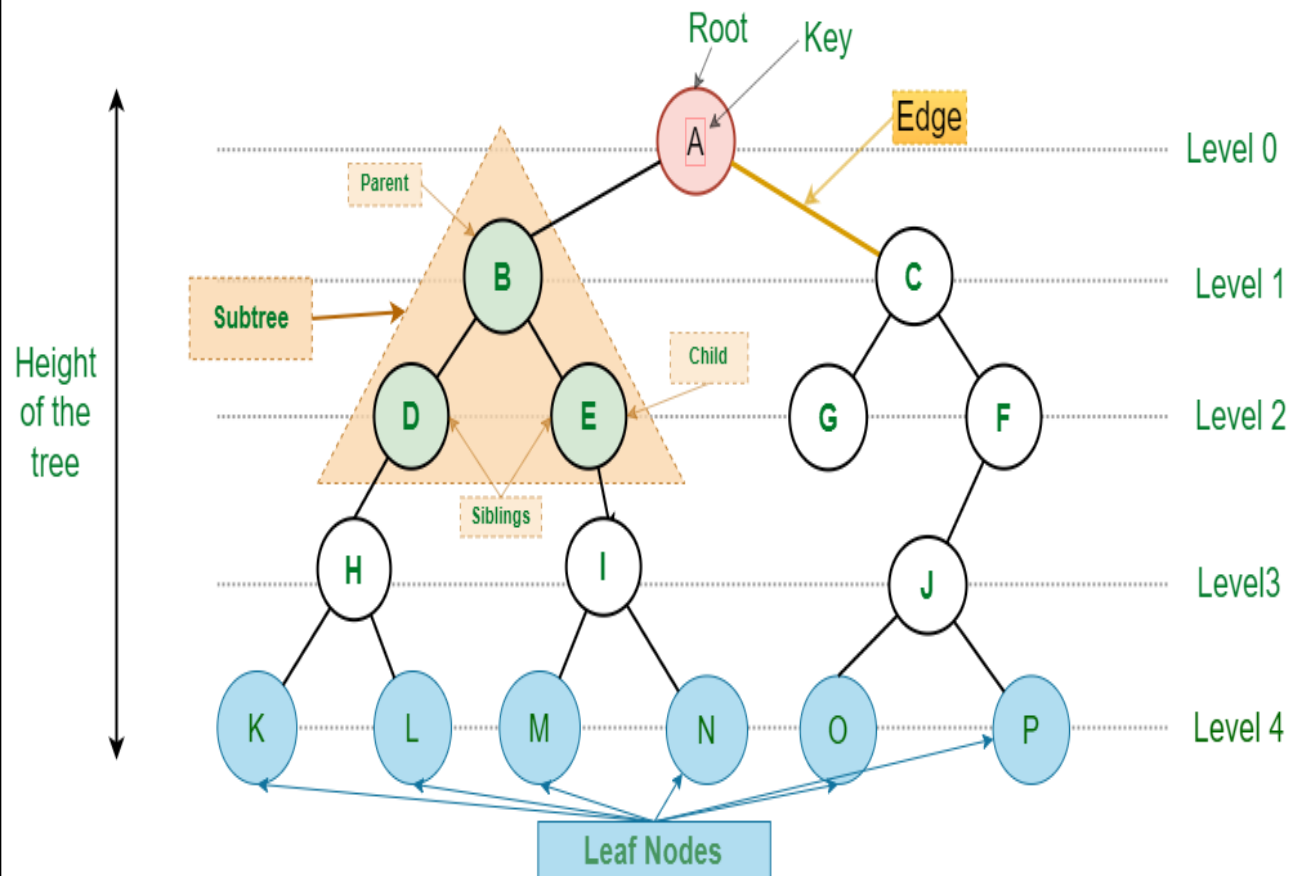
- A tree is a **non-linear data structure made up of nodes or vertices and edges which is organized in hierarchical structure (without having any cycle)**
- In tree data structure, every individual element is called as **Node**. Node in a tree data structure stores the actual data of that particular element and link to next element in hierarchical structure.
- Collection of nodes that are connected by edges and has a hierarchical relationship between the nodes.
- In a tree data structure, if we have **N** number of nodes then we can have a maximum of **N-1** number of links.
- The **tree with no nodes is called the null or empty tree.**
- A tree that is not empty consists of a root node and potentially many levels of additional nodes that form a hierarchy.



TREE with 11 nodes and 10 edges

- In any tree with '**N**' nodes there will be maximum of '**N-1**' edges
- In a tree every individual element is called as '**NODE**'

Tree Data Structure

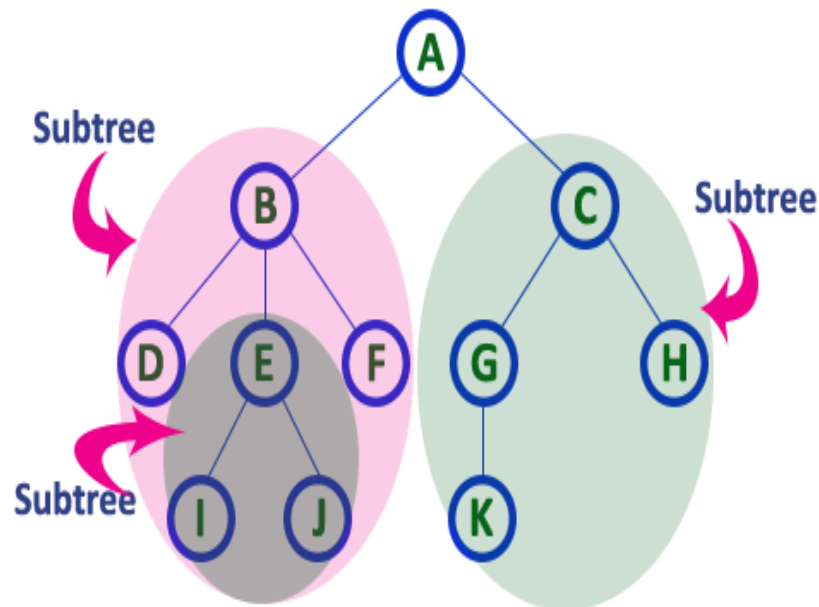


Terminology

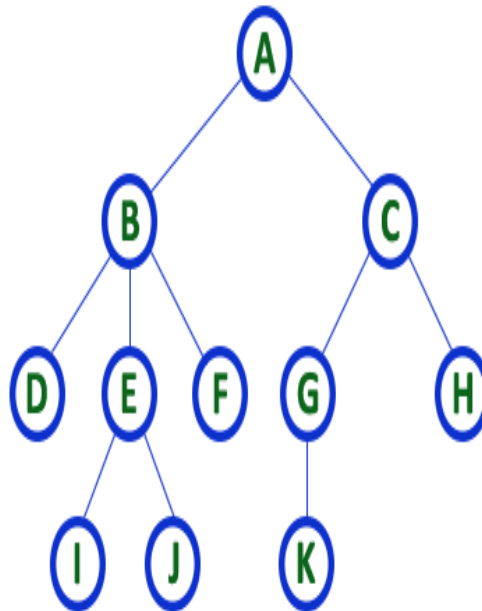
- **Node:** A node is an entity that contains a key or value and pointers to its child nodes.
- **Edge:** It is the link between any two nodes. The reference through which a parent node is connected to a child node is called an edge.
- **Root node:** The topmost node of a tree or the node which does not have any parent node is called the root node. It is the root node of the tree. A non-empty tree must contain exactly one root node and exactly one path from the root to all other nodes of the tree.
Ex.: {A} is the root node of the tree.
- **Parent Node:** The node which is a predecessor of a node is called the parent node of that node. {B} is the parent node of {D, E}.
- **Child Node:** The node which is the immediate successor of a node is called the child node of that node. Examples: {D, E} are the child nodes of {B}.

- **Leaf Node or External Node:** The nodes which do not have any child nodes are called leaf nodes. {K, L, M, N, O, P, G} are the leaf nodes of the tree.
- **External Node:** The node with no child node are called as External node. Leaf nodes are also called as **External Nodes**.
- **Internal node:** A node with at least one child is called Internal Node. Are also called as Non leaf node.
- **Ancestor of a Node:** Any predecessor nodes on the path of the root to that node are called Ancestors of that node. {A,B} are the ancestor nodes of the node {E}
- **Descendant:** Any successor node on the path from the leaf node to that node. {E,I} are the descendants of the node {B}.
- **Sibling:** Children of the same parent node are called siblings. {D,E} are called siblings.

- **Neighbour of a Node:** Parent or child nodes of that node are called neighbors of that node.
- **Subtree:** Any node of the tree along with its descendant.



Degree: the total number of children of a node is called as degree of that Node. In simple words, the Degree of a node is total number of children it has. The highest degree of a node among all the nodes in a tree is called as 'Degree of Tree'.



Here Degree of B is 3

Here Degree of A is 2

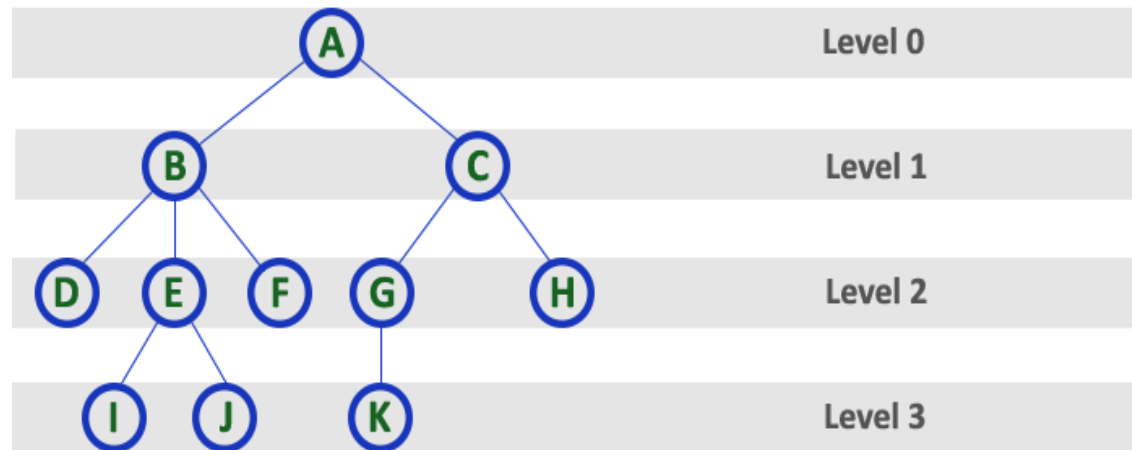
Here Degree of F is 0

- In any tree, 'Degree' of a node is total number of children it has.

Level of a node: The count of edges on the path from the root node to that node.

In a tree each step from top to bottom is called as a Level and the Level count starts with '0 ' and incremented by one at each level (Step).

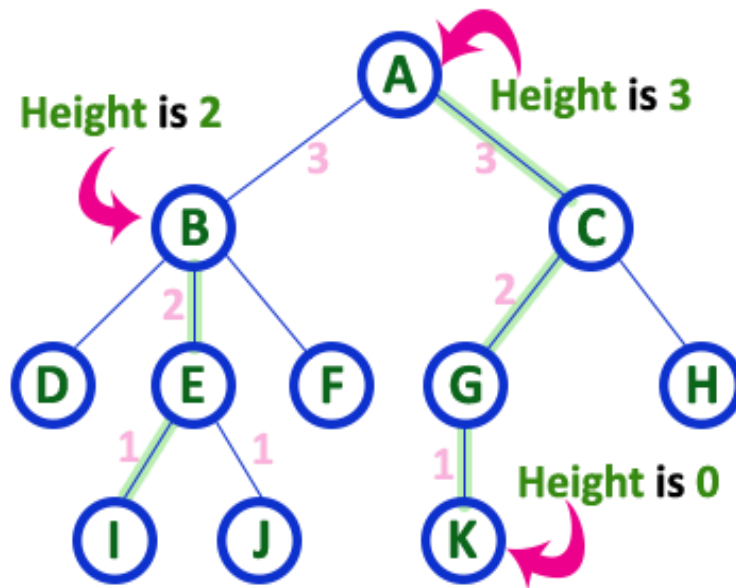
The root node is said to be at Level 0 and the children of root node are at Level 1 and the children of the nodes which are at Level 1 will be at Level 2 and so on...



Height of the node:

In a tree data structure, the total number of edges from leaf node to a particular node in the longest path is called as HEIGHT of that Node.

height of the tree: In a tree, height of the root node is said to be **height of the tree**. In a tree, **height of all leaf nodes is '0'**.



Here Height of tree is 3

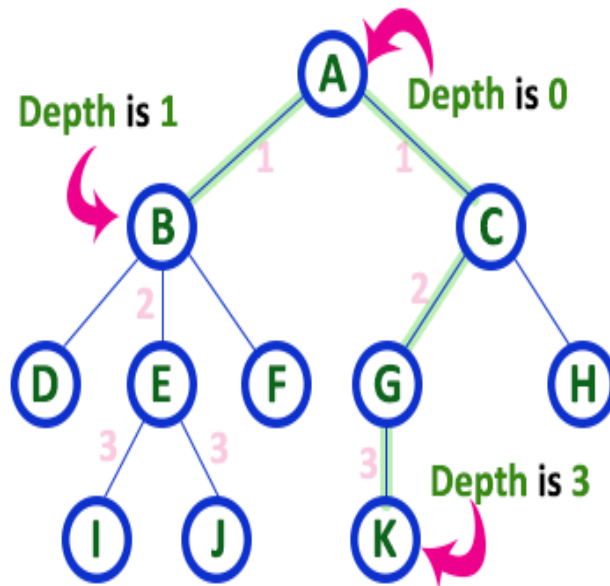
- In any tree, 'Height of Node' is total number of Edges from leaf to that node in longest path.
- In any tree, 'Height of Tree' is the height of the root node.

Depth

In a tree data structure, the total number of edges from root node to a particular node is called as **DEPTH** of that Node.

In a tree, the total number of edges from root node to a leaf node in the longest path is said to be **Depth of the tree**.

In simple words, the highest depth of any leaf node in a tree is said to be depth of that tree. In a tree, **depth of the root node is '0'.**



Here Depth of tree is 3

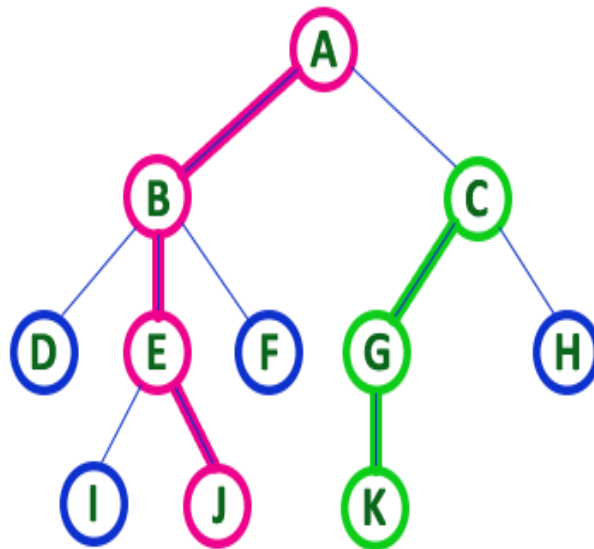
- In any tree, 'Depth of Node' is total number of Edges from root to that node.
- In any tree, 'Depth of Tree' is total number of edges from root to leaf in the longest path.

Path

In a tree data structure, the sequence of Nodes and Edges from one node to another node is called as **PATH** between that two Nodes.

Length of a Path is total number of nodes in that path.

Example **the path A - B - E - J has length 4.**



- In any tree, 'Path' is a sequence of nodes and edges between two nodes.

Here, 'Path' between A & J is

A - B - E - J

Here, 'Path' between C & K is

C - G - K